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Bwrdd Iechyd Prifysgol Betsi Cadwaladr University Health Board

Optimisation of Anaemia following Major Bowel Surgery

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1. Introduction

1.1	The aim of the document is to guide healthcare professionals in screening patients for and optimising patients with post operative anaemia who have undergone surgery thereby reducing post operative transfusion requirements and supporting patient recovery.
1.2	Anaemia is defined according to WHO criteria[1] as Haemoglobin (Hb) concentration <120g/L in women and <130g/L in men.
1.3	Iron deficiency can be defined[2] by: Ferritin concentration <100mcg/L or Ferritin <300mcg/L and Transferrin Saturation <20% or Reticulocyte Hb content <28pg.
1.4	Both pre-operative and post-operative anaemia have been identified as independent risk factor for complications and death[5].
1.5	Anaemia may be worse in the post-operative period due to critical illness, reduced erythropoietin production and secretion, decreased iron absorption from the GI tract and impaired mobilisation from body stores.
1.5	Reduction in the use of red cells for anaemia management is cost effective and conserves blood supplies for those patients who need it most. There is also evidence that transfusion of blood is associated with increased morbidity and mortality[6].

2. Scope

2.1	This document applies to any patient who has undergone non-day case, major elective or emergency bowel surgery
2.2	A major surgical procedure is defined, in this context, as any procedure for which a group and screen or cross match is recommended. (See BCUHB Blood Transfusion Requirements document)
2.3	This document is intended for all physicians, nurses and pharmacists caring for patients in the post operative period.
2.4	This document complements the Pan-BCU guideline for iron deficiency anaemia in adults.

3. Responsibilities

3.1	**Ward Nurse**
	Identify patients who are undergoing major surgery.
	Ensure cannula flushed with 10mls of 0.9% normal saline and patent prior to administration of intravenous iron.
	Ensure regular review of the cannula to check it remains patent and has not tissued during infusion of intravenous iron.
	Order intravenous iron once prescribed by doctor and ensure given in a timely fashion
	Liaise with pharmacist if any treatment is required (vitamin B12, folate, oral and

	intravenous iron, epo).
	Provide patient information and explain the benefits and risks of therapy offered.
	Safe administration of intravenous medications. Trained in evaluation and management of Anaphylaxis.

3.2	**Surgical team**
	Identify patients who are undergoing major surgery.
	Check Hb daily until at least the third day post operatively.
	Measure and review Ferritin level and Transferrin saturations on the day following surgery.
	Send additional 'anaemia' tests when appropriate.
	Review blood test results no later than the next working day.
	Liason with pharmacist and/or consultant surgeon to agree on plan of care.
	Prescribe intravenous iron therapy and make ward nurse aware.
	Liaise with ward pharmacist, if advice is required.
	Provide patient information and explain the benefits and risks of therapy offered.
	Familiar with identification and management of Anaphylaxis.
	Inform Primary care team of any iron treatment the patient has received.
	Arrange and review post therapy blood tests.

3.3	**Patient/Carer**
	Understand the need for investigation and treatment of anaemia.
	Understand the benefits and risks of intravenous therapy.
	Inform ward nurse if pain on injection with fluids when cannula checked and flushed.

4. Guidance

4.1	**Patient Assessment**
	Haemoglobin should be monitored daily in major surgery, until at least the third post-operative day.
	Measure Ferritin level & Transferrin saturations +/- Reticulocyte Hb content on the day after surgery. N.B Ferritin will not be increased by inflammation immediately following surgery which is why it can be measured to aid the diagnosis of iron deficiency anaemia.
	See Appendix 1 for diagnosis of 'iron deficiency'

4.2	**Interventions**
	If iron replacement is indicated (Appendix 1), it is recommended to give IV iron transfusion early.
	If patient has moderate to severe postoperative anaemia (Hb <110g/L), they may benefit from IV iron transfusion.
	If severe post-operative anaemia occurs despite actions to avoid this, it is

	recommended to adopt a restrictive transfusion threshold – accepting a Hb of 70-80g/L depending on patient comorbidities such as cardiovascular disease.

4.3	**Intravenous Iron Therapy (Ferinject: Ferric Carboxymaltose)**
	Indication: - See Appendix 1
	Benefits: • Due to elevated hepcidin levels following major surgery, oral iron supplements are of limited efficacy. • IV iron does not increase rates of infection post operatively, whereas red cell transfusion delivers both haem and labile iron which supports bacterial cell growth.
	Avoid in the first trimester of pregnancy.
	Expected effect on Hb: A rapid Hb response is exhibited in most patients (50% at 5 days, 75% at 10–14 days, maximal at 3 weeks).
	Stop any oral iron prescription
	Side Effects: • Allergic reaction (<1%) • Permanent skin staining (Rare). Ensure check patency of cannula prior to starting intravenous iron infusion. • Headache, dizziness, rash, nausea, abdominal discomfort, muscle cramps, flushing (1-10%) • Injection site reactions – erythema, swelling, burning, pain, bruising, discolouration, extravasation, irritation, reaction.
	Prescribing & Administration: See Appendix 2
	After Administration: Repeat FBC two to three weeks after administration. Inform GP and ask them to check FBC in the community, if the patient is discharged prior to three weeks post iron transfusion.

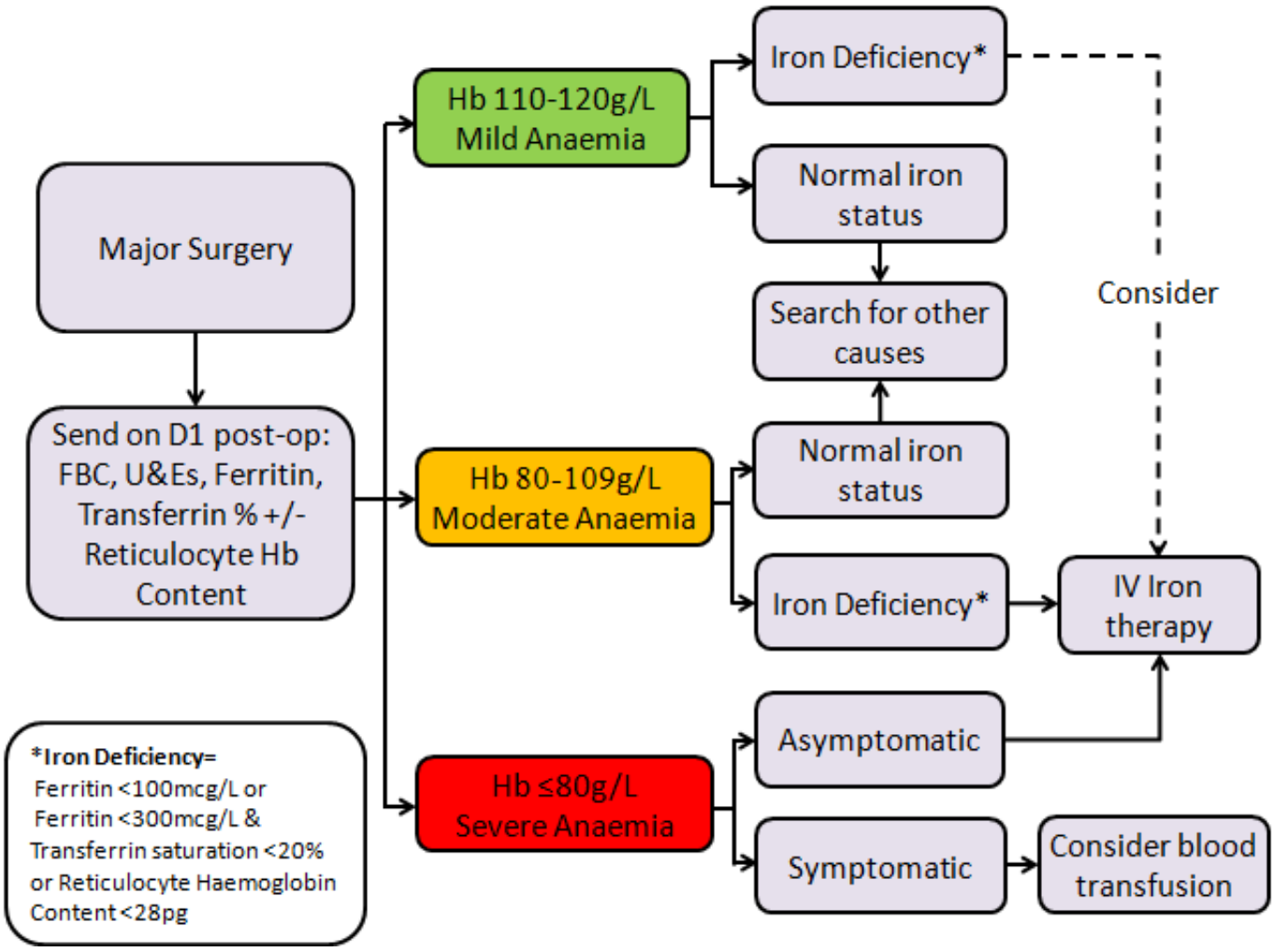
4.4	**Patient information**
	Patients with capacity should be provided with the following information in order to gain valid verbal consent : - rationale for blood investigations and recommended treatment. - Treatment side effects. - Copy of Appendix 3. If a patient is unable to provide verbal consent due to lack of capacity, information should be provided to relatives and this discussion documented in patient notes.

5. References

1. International consensus statement on the peri-operative management of anaemia and iron deficiency. *Anaesthesia* 2017, 72, 233–247
2. M. Monaz et al. An international consensus statement on the management of postoperative anaemia after major surgical procedures. *Anaesthesia* 2018, 73, 1418–1431.
3. Shander A, Knight K, Thurer R, et al. Prevalence and outcomes of anemia in surgery: a systematic review of the literature. *Am J Med* 2004; 116 (Suppl.7A):58S–69S.
4. Musallam KM, Tamim HM, Richards T, et al. Preoperative anaemia and postoperative outcomes in noncardiac surgery: a retrospective cohort study. *Lancet* 2011; 378:1396–1407.
5. Ferraris VA, Davenport DL, Saha SP, Austin PC, Zwischenberger JB.99–604 Surgical outcomes and transfusion of minimal amounts of blood in the operating room. *Arch Surg*. 2012;147(1):49-55.
6. Glance LG, Dick AW, Mukamel DB et al. Association between intraoperative blood transfusion and mortality and morbidity in patients undergoing non-cardiac surgery. *Anesthesiology* 2011; 114:283-92

6. Appendices

Appendix 1: Post operative anaemia management (adapted from reference 2)



Appendix 2: Ferinject Prescribing & Administration

FERINJECT

ENSURE ANAPHYLAXIS KIT AND EMERGENCY TROLLEY ARE AT HAND AND IN DATE BEFORE STARTING ANY INFUSION.

CHECK THE CANNULA IS NOT TISSUED PRIOR TO USE

Presentation

Inspect all vials for sediment and damage before use. Only use if it appears to be a sediment-free homogenous solution. Each vial is for single use only.

Volume	Iron Content
10mL	500mg
20mL	1000mg

Determination of the Dose

Patient Body Weight	
<35 kg	**≥35 kg**
500 mg	1000mg

Infusion Preparation

Ferinject must only be diluted with Sodium Chloride 0.9% solution.

Ferinject (50mg/mL)	Iron	Maximum Volume of Sodium Chloride 0.9%	Minimum Administration Time
10-20mL	500-1000mg	250mL	15 minutes

Maximum single dose in one week = 1000mg (1g)

Observations

Monitor and record NEWS vital signs prior to infusion, 5 minutes after commencing and at the end of the infusion. Continue to monitor until 30 minutes after completion of infusion for signs of allergic reaction.

Adverse Reactions

If any adverse reactions are noted at any time throughout the infusion:

1. STOP the infusion immediately
2. Contact for medical assistance
3. Contact Resuscitation team in the event of cardiac arrest, anaphylactic shock or respiratory distress.

Extravasation

In the event of extravasation, the following steps should be carried out:

- Check venflon site is correct
- Aspirate any residue from the cannula site
- Flush the cannula with sodium chloride 0.9%, then remove cannula.

Document the event in the medical notes & inform the prescriber.

Reference: Summary of Product Characteristics Ferinject. Last updated 9 July 2019. Available at: https://www.medicines.org.uk/emc/medicine/23669. Accessed 28/8/19

Appendix 3: Patient Information

Taflen wybodaeth therapi haearn i mewn i wythïen (Mewnwythiennol)

Wedi i ni brofi eich gwaed ar ol eich llawdriniaeth, dangosodd bod lefelau haearn yn eich gwaed yn isel. Gelwir hyn yn anaemia.

Mae hi yn bwysig cael digon o haearn yn eich gwaed er mwyn hybu eich corff i wella yn dilyn y llawdriniaeth. Gellir lefelau isel o haearn achosi symptomau o fod yn flinedig ac yn fyr o wynt.

Mae eich meddyg wedi penderfynu rhagnodi triniaeth haearn (Ferinject), Pigiad uniongyrchol i wythïen yn eich braich yw Ferinject. Nid yw tabledi haeran yn cael eu amsugno yn effeithiol ar ol llawdriniaeth.

Bydd y profion gwaed yn cael eu ailadrodd 2-3 wythnos ar ol y driniaeth er mwyn sicrhau fod lefel y haearn yn y gwaed wedi codi.

Mae Ferinject yn cael ei ystyried yn saff. Mae adwaith alergedd yn gallu digwydd on yn anaml (llai na 1%) bydd y nyrs yn eich monitro yn agos yn ystod ac arol y chwistrelliad.

Mae cymlethdodau arwyddocaol gyda Ferinject, ond maent yn brin. Un ohonynt yw newid lliw parhaol o'r croen. Mae hyn yn gallu digwydd os yw'r meddiginiaeth yn gollwing tu allan i'r wythien. Er mwyn lleihau y risg bydd y wythien yn cael ei golchi gyda dwr neu saline (dwr hallt) cyn y chwistrelliad.

Mae sgil effeithiau eraill yn cynwys, cyr pen, pendro,brech ar y croen, cyfog a thaflu I fynu, poen abdomenol, chramp y cyhyra, dolur rhydd neu bod yn rhwym, effaith ar y iau, gwrid, pwysedd gwaed uchel neu isel ac adwaith ar safle y chwistrelliad.

Iron therapy into a vein (Intravenous) information leaflet.

We tested your blood after your operation and it showed that the iron level in your blood was low. This is known as anaemia.

It is important to have enough iron to help your body recover from the operation. Low iron can give you symptoms of feeling tired and breathless.

Your doctor has decided to prescribe iron treatment (Ferinject). This will be an injection you will receive directly into a vein in your arm. This is because oral iron is not well absorbed after surgery.

We will re-test your bloods 2-3 three weeks after you receive the iron injections to make sure your levels have risen.

Ferinject is considered safe to use. Rarely, (in less than 1% of cases) it can cause allergic reactions. Therefore, you will be monitored closely before, during and after the infusion.

A rare, but significant complication of Ferinject is permanent skin staining or discoloration, which can occur if some of the drug leaks outside the vein during the infusion. To reduce the risk, a flush of water or saline (salt solution) is given into your vein before the Ferinject.

Potential mild side effects occur in 1 to 10% of patients, including headache, dizziness, rash, nausea and vomiting, abdominal pain, muscle cramps, diarrhoea, constipation, abnormal liver function, flushing, low or high blood pressure and injection site reactions.

7. Members of the Working Group

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